

Ashitosh Kisan Jagtap

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SUMMARY

Full-stack developer and ML engineer with 6 months of development experience including 3 freelance client projects. Architected AI-powered security platforms, real-time competitive coding applications, and multi-language audio classification systems. National finalist (Top 20 of 40,000) at HCL GUVI AI Impact Buildathon. Proficient in Python, React, Node.js, Express, FastAPI, PyTorch, MongoDB, and Docker.

EXPERIENCE

Freelance Full-Stack Developer

Oct 2025 - Present

- Delivered 3 end-to-end client projects handling requirements gathering, system design, full-stack development, testing, and cloud deployment using React, Node.js, FastAPI, and Docker.
- Engineered real-time systems with WebSocket (Socket.io) achieving sub-200ms latency, including live matchmaking, sandboxed code execution, and streaming threat analysis.
- Designed and documented RESTful APIs with JWT authentication, bcrypt password hashing, express-rate-limit middleware, and role-based access control serving 10+ endpoints per project.
- Deployed containerized production applications on cloud infrastructure using Docker with MongoDB, Redis, PostgreSQL, and SQLite database backends.

PROJECTS

SafeWaves - AI-Powered Multi-Threat Cyber Defense Platform

safewaves.vercel.app

Python, React, FastAPI, PyTorch, Tailwind CSS, Zustand, SQLite

- Engineered a multi-threat cyber defense platform detecting phishing, malicious URLs, deepfakes, prompt injection, behavioral anomalies, and AI-generated content across 6 distinct threat categories.
- Implemented explainable ML pipeline with risk scoring, SHAP-based feature attribution, natural-language threat explanations, and automated remediation recommendations for each detected threat.
- Architected React frontend with Tailwind CSS and Zustand state management, integrated with FastAPI backend using Server-Sent Events (SSE) for real-time threat streaming and SQLite persistence.
- Developed adversarial testing suite evaluating model robustness against 500+ obfuscated malicious input variations, reducing false positive rate by 15% through iterative model tuning.

CodeBattle - Real-Time Competitive Coding Platform

Node.js, React, MongoDB, Socket.io

Node.js, Express, React, MongoDB, Socket.io, Monaco Editor, JWT, VM2

- Built a full-stack competitive coding platform where developers battle head-to-head solving algorithmic challenges in real-time with live opponent progress tracking via WebSocket.
- Implemented WebSocket-based matchmaking engine with ELO rating system (dynamic K-factor), matching players within 200 rating points in under 2 seconds across difficulty tiers.
- Engineered sandboxed server-side JavaScript code execution using VM2 with 5-second timeout protection, running user submissions against visible and hidden test cases with detailed feedback.
- Designed 12+ RESTful endpoints with JWT auth, bcrypt hashing, paginated leaderboards, match history, and user profiles; React frontend with Monaco Editor and split-pane battle arena.

HONORS & AWARDS

National Finalist, Top 20 of 40,000 Participants

Feb 2026

HCL GUVI AI Impact Buildathon

- Engineered an end-to-end ML pipeline for AI-generated voice detection achieving 94%+ classification accuracy across 5 languages using PyTorch, librosa audio feature extraction, and custom CNN architecture.

Top 30 Finalist, 300+ Teams

Mar 2026

IndiaNext Hackathon 2026, K.E.S. Shroff College, Mumbai

- Developed an AI-powered cybersecurity platform integrating 6 threat detection modules with explainable AI, real-time risk scoring, and automated incident response recommendations.

EDUCATION

Indian Institute of Technology (IIT) Madras

2023 - 2027

Bachelor of Science (BS), Data Science and Applications

Relevant Coursework: Deep Learning, Machine Learning, Reinforcement Learning, Data Structures & Algorithms, Database Systems, Statistics

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, HTML, CSS

Frameworks: React, Node.js, Express.js, Flask, FastAPI, REST APIs, Socket.io, Tailwind CSS

ML / AI: PyTorch, Scikit-learn, XGBoost, LLMs, SHAP (Explainable AI), Audio Processing, CNNs

Databases: MongoDB, PostgreSQL, SQLite, Redis

DevOps / Tools: Docker, Git, GitHub, Linux, Vercel, VS Code, Postman, npm

Concepts: Machine Learning, Deep Learning, NLP, REST API Design, Microservices, CI/CD, WebSocket, Agile, Cybersecurity

ADDITIONAL

- Attended AI Impact Summit India 2026; engaged with 50+ AI startups on MLOps pipelines and scalable AI deployment strategies.
- Active open-source contributor on GitHub with documented repositories, version-controlled workflows, and comprehensive project documentation.